

Synthetic Cultural Agents via Structural Preference Estimation: DPO as Bradley–Terry MLE

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Abstract.^a We demonstrate that the Direct Preference Optimization (DPO) training objective is a reparameterization of the Bradley–Terry (BT) log-likelihood, where latent utility differences are expressed as policy log-probability ratios relative to a reference distribution. This reparameterization equivalence - not merely an analogy - allows us to import the full toolkit of structural econometrics (identification, overidentification testing, counterfactual analysis) into the preference-based fine-tuning of large language models. We leverage this result to formalize the *Synthetic Cultural Agent* (SCA 2.0) framework: a structurally grounded pipeline that calibrates language model behavior to non-WEIRD populations using a culture-specific reward function parameterized by the Global Preferences Survey (GPS) vector. We propose a hierarchical moment-selection strategy - identifying moments from GPS, overidentifying moments from the World Values Survey (WVS), and held-out behavioral tasks - that provides formal tests of structural validity via the J -statistic. We illustrate the complete pipeline with an end-to-end worked example for Argentina, including teacher-student distillation, implied reward recovery, three-tier validation against WVS Wave 7, and counterfactual comparative statics.

^aThis working paper is for internal distribution only at the moment. The goal is to work out a known theoretical connections between Direct Preference Optimization (DPO) and the MLE estimator of the Bradley-Terry (BT) discrete choice model, and establish a research agenda to improve the methodology used in the Synthetic Cultural Agents (SCA) project.

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1. Introduction: The Structural Claim

The rapid ascent of Large Language Models (LLMs) presents a paradox for the structural economist. These models appear to be atheoretical “black boxes,” yet they exhibit sophisticated capabilities to reason on Language space. This paper argues that the current frontier of AI alignment - specifically Direct Preference Optimization (DPO) (Rafailov, Sharma, et al. 2023) - is not an ad-hoc engineering heuristic, but a rigorous application of **Revealed Preference Theory** and **Stochastic Control**.

We evaluate the following precise statement:

Direct Preference Optimization (DPO) is equivalent to maximum-likelihood estimation in a Bradley–Terry random utility model, where the latent utility is structurally identified as the log-probability ratio relative to a reference policy.

“Equivalent” here means: the DPO training objective *is* the Bradley–Terry log-likelihood for a particular score function. It does *not* mean that all RLHF variants reduce to BT, nor that BT perfectly models human judgment noise. The point is methodological: it is legitimate to import structural-econometric language (utility, revealed preference, likelihood, identification) into preference tuning with minimal translation costs.

This paper makes three contributions:

1. **The reparameterization equivalence.** We prove that DPO policy estimation and BT MLE are the same optimization problem (Section 2). The proof is straightforward - it requires only the closed-form solution to a KL-regularized control problem and algebraic substitution - but the implications for both fields are nontrivial. We state the assumptions explicitly (A1–A4), highlight what is and is not identified, and add remarks on the exogeneity of β and the inexact Hotz–Miller analogy.
2. **The SCA 2.0 framework with formal validation.** We define Synthetic Cultural Agents structurally as policies maximizing a culture-specific reward function parameterized by the Global Preferences Survey (GPS) vector $z \in \mathbb{R}^6$ (Falk et al. 2018). We formalize the current WVS-comparison validation as a Simulated Method of Moments (SMM) procedure and propose a hierarchical moment-selection strategy with an overidentification test (Section 3).
3. **An end-to-end application.** We walk through the complete SCA pipeline for Argentina: from GPS calibration through teacher-student distillation, implied reward recovery, three-tier validation against WVS Wave 7, and counterfactual comparative statics (Section 5).

This work builds on an earlier approach to cultural modeling of LLMs. In Gonzalez-Bonorino, Capra, and Pantoja (2025) - hereafter *SCA 1.0* - Synthetic Cultural Agents were constructed via **persona prompting** (also called “profiling”): a frontier LLM was conditioned on a demographic and cultural profile at inference time, shifting the *reference distribution* through in-context information without modifying the model’s parameters. SCA 1.0 demonstrated that persona-prompted LLMs can reproduce non-WEIRD behavioral patterns in economic experiments, but the approach has two

fundamental limitations: 1) the cultural signal lives entirely in the prompt context and vanishes when the context window is exhausted or the model is restarted, and 2) results are highly sensitive to model and prompt specifications, making the methodology fragile to model updates. In contrast, *SCA 2.0* approaches the task through the lens of **structural estimation**: a culture-specific reward function $R(x, y; z)$ is used to DPO-fine-tune the model, engraining cultural preferences directly into the model’s weights. The shift from profiling to structural estimation - from shifting the reference distribution to optimizing a reward function - is the methodological advance of this paper.

A method is *structural* if it (i) posits economic primitives (preferences, technology, information), (ii) derives testable implications (choice probabilities, moments), and (iii) recovers primitives from data via identification arguments. DPO satisfies all three criteria: the primitives are the latent reward r^* and the reference policy π_{ref} ; the testable implications are the BT preference probabilities; and identification proceeds via the log-ratio inversion. In contrast, supervised fine-tuning directly fits $\pi_{\theta}(y|x)$ to observed (x, y) pairs without a behavioral model.

Roadmap. Section 2 proves the equivalence. Section 3 develops the moment-selection theory. Section 4 defines the *SCA 2.0* framework. Section 5 presents the Argentina application. Section 6 extends the analysis to sequential decisions. Section 7 discusses limitations. Appendix A collects proofs; Appendix B presents a side-by-side derivation from the LLM and econometrics literatures; Appendices C and D provide a dictionary and a worked micro-example.

2. The Equivalence: DPO as Bradley–Terry MLE

2.1. Preference Data and the Bradley–Terry Model

Let $x \in \mathcal{X}$ denote a context (prompt, task, environment). We observe preference comparisons

$$\mathcal{D} = \{(x_i, y_i^{(w)}, y_i^{(\ell)})\}_{i=1}^N,$$

where $y_i^{(w)}$ is judged preferred to $y_i^{(\ell)}$ given x_i .

Definition 1 (Bradley–Terry model). Given a real-valued score (latent utility / reward) $r^* : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, the BT model specifies paired-comparison probabilities as

$$\mathbb{P}(y_1 \succ y_2 \mid x) = \sigma(r^*(x, y_1) - r^*(x, y_2)), \quad (1)$$

where $\sigma(u) = (1 + e^{-u})^{-1}$ is the logistic sigmoid.

This is the paired-comparison specialization of the logit random utility model (McFadden 1974; Bradley and Terry 1952).

Theorem 1 (Luce–McFadden microfoundation). *Let latent utility be $U(x, y) = r^*(x, y) + \varepsilon_y$, where $\{\varepsilon_y\}_{y \in \mathcal{Y}(x)}$ are i.i.d. Type-I Extreme Value (Gumbel) random variables. Then*

$$\mathbb{P}(y_1 \succ y_2 \mid x) = \mathbb{P}(U(x, y_1) \geq U(x, y_2)) = \sigma(r^*(x, y_1) - r^*(x, y_2)).$$

Proof sketch. The difference of two i.i.d. Gumbel random variables has a logistic distribution: $\mathbb{P}(\varepsilon_2 - \varepsilon_1 \leq z) = \sigma(z)$. Apply with $z = r^*(x, y_1) - r^*(x, y_2)$. $\square$

Given labeled pairs $\mathcal{D}$, the BT log-likelihood is

$$\ell(r; \mathcal{D}) = \sum_{i=1}^N \log \sigma(r(x_i, y_i^{(w)}) - r(x_i, y_i^{(\ell)})). \quad (2)$$

Proposition 1 (Identification up to location). *For each fixed x , the reward function $r^*(x, \cdot)$ is identified only up to an additive constant $c(x)$, since preference probabilities depend only on differences.*

Remark 1 (Why Gumbel matters). The Gumbel assumption yields the log-sum-exp structure that appears throughout: $\mathbb{E}[\max_y \{r(x, y) + \varepsilon_y\}] = \beta \log \sum_y \exp(r(x, y)/\beta) + \text{const}$. This is the “inclusive value” in discrete choice (McFadden 1974) and reappears as the value function in soft RL (Section 6).

2.2. The Planner’s Problem (RLHF-KL)

The alignment problem is a KL-regularized control problem. We seek a policy π that maximizes expected reward while remaining close to a reference policy π_{ref} (the pre-trained LLM):

$$\max_{\pi(\cdot|x)} \mathbb{E}_{y \sim \pi(\cdot|x)} [r^*(x, y)] - \beta D_{\text{KL}}(\pi(\cdot|x) \parallel \pi_{\text{ref}}(\cdot|x)), \quad (3)$$

where $\beta > 0$ is the regularization parameter (temperature).

Theorem 2 (Closed-form optimal policy). *Assume $r^*(x, \cdot)$ is bounded and $\pi_{\text{ref}}(y | x) > 0$ on the support of interest. The objective (3) is strictly concave in $\pi(\cdot | x)$, and the unique optimal policy is the tilted Gibbs distribution:*

$$\pi^*(y | x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y | x) \exp\left(\frac{r^*(x, y)}{\beta}\right), \quad (4)$$

where $Z(x) = \sum_{y'} \pi_{\text{ref}}(y' | x) \exp(r^*(x, y')/\beta)$ is the partition function.

Proof. See Appendix A.3. □

Corollary 1 (Log-ratio identity). *Let π^* satisfy Theorem 2. Then for any $y_1, y_2 \in \mathcal{Y}(x)$,*

$$r^*(x, y_1) - r^*(x, y_2) = \beta \left[\log \frac{\pi^*(y_1 | x)}{\pi_{\text{ref}}(y_1 | x)} - \log \frac{\pi^*(y_2 | x)}{\pi_{\text{ref}}(y_2 | x)} \right]. \quad (5)$$

The normalizer $\beta \log Z(x)$ cancels in differences.

Key Result

The Inversion Formula. Reward differences can be expressed as log-ratios of the optimal policy relative to the reference, without needing to compute the intractable partition function $Z(x)$. This is the structural inversion that makes DPO possible.

2.3. Assumptions

We isolate the exact conditions under which “DPO is BT MLE” holds.

- (A1) **BT data-generating process.** Conditional on (x, y_1, y_2) , the probability that a labeler prefers y_1 to y_2 is $\sigma(r^*(x, y_1) - r^*(x, y_2))$ for some latent r^* .
- (A2) **KL-optimality relation.** The policy that “implements” those rewards is the optimizer of the KL-regularized planner (3) with reference π_{ref} and temperature β , giving the tilted Gibbs form (4).
- (A3) **Support compatibility.** $\pi_{\text{ref}}(y | x) > 0$ wherever the policy family (and the data) place mass, so that log-ratios are well-defined.
- (A4) **Estimation family.** We fit a parametric policy $\pi_\theta(\cdot | x)$ by maximum likelihood under the implied model. Correct specification ($\exists \theta^* : \pi_{\theta^*} = \pi^*$) is *not* needed for objective equivalence, only for “recover the true policy” statements.

2.4. Main Result

Proposition 2 (DPO objective equals BT log-likelihood for log-ratio scores). *Under (A1)–(A4), the BT likelihood for preference data can be written as a function of policy log-ratios:*

$$\mathbb{P}(y^{(w)} \succ y^{(\ell)} \mid x) = \sigma \left(\beta \left[\log \frac{\pi^*(y^{(w)} \mid x)}{\pi_{\text{ref}}(y^{(w)} \mid x)} - \log \frac{\pi^*(y^{(\ell)} \mid x)}{\pi_{\text{ref}}(y^{(\ell)} \mid x)} \right] \right).$$

Replacing π^* by a parametric family π_θ and maximizing the resulting log-likelihood yields the DPO estimator:

$$\hat{\theta}_{\text{DPO}} \in \arg \max_{\theta} \sum_{i=1}^N \log \sigma \left(\beta \left[\log \frac{\pi_\theta(y_i^{(w)} \mid x_i)}{\pi_{\text{ref}}(y_i^{(w)} \mid x_i)} - \log \frac{\pi_\theta(y_i^{(\ell)} \mid x_i)}{\pi_{\text{ref}}(y_i^{(\ell)} \mid x_i)} \right] \right). \quad (6)$$

The DPO loss $\mathcal{L}_{\text{DPO}}(\theta) = -\ell(\theta; \mathcal{D})$ used in practice is the negative of this log-likelihood.

Proof sketch. The argument proceeds in five steps:

1. **Invert the policy.** From Theorem 2: $r^*(x, y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x)$.
2. **Compute reward differences.** For any pair $(y^{(w)}, y^{(\ell)})$:

$$r^*(x, y^{(w)}) - r^*(x, y^{(\ell)}) = \beta \left[\log \frac{\pi^*(y^{(w)} \mid x)}{\pi_{\text{ref}}(y^{(w)} \mid x)} - \log \frac{\pi^*(y^{(\ell)} \mid x)}{\pi_{\text{ref}}(y^{(\ell)} \mid x)} \right].$$

The partition function $Z(x)$ cancels. This is the crucial step.

3. **Substitute into BT.** Using (A1), $\mathbb{P}(y^{(w)} \succ y^{(\ell)} \mid x) = \sigma(\beta \Delta \log(\pi^*/\pi_{\text{ref}}))$.
4. **Form the log-likelihood.** Sum log of the above across observations $i = 1, \dots, N$.
5. **Replace π^* with π_θ .** Since π^* is unknown, posit a parametric family and maximize, yielding (6). $\square$

A line-by-line derivation with all intermediate steps is given in Appendix A.4.

2.5. Remarks and Caveats

Remark 2 (β is exogenous). The temperature parameter β is fixed by the practitioner, not recovered from data. In the structural interpretation, this means the scale of the latent utility is not identified separately from β . In the econometrics literature, the scale parameter in logit models is typically estimated jointly with the utility parameters (or normalized to unity). Extending DPO to jointly estimate β is an open problem; see Section 7.

Remark 3 (The Hotz–Miller analogy is suggestive but not exact). DPO inverts a *posited optimal policy* for one-period rewards, while Hotz–Miller (Hotz and Miller 1993) inverts *observed conditional choice probabilities* (CCPs) for choice-specific value functions that include continuation values in a dynamic setting. The algebra is the same - both exploit the closed-form relationship between logit

choice probabilities and utility differences - but the objects differ. In DPO, π_{ref} is a known reference distribution; in DDC, the base measure is determined by the shock distribution. The DPO setting is static (one-step); extending to sequential decisions requires the soft Bellman framework (Section 6).

Remark 4 (Misspecification). If the BT model is misspecified (e.g., preferences violate IIA-like properties, raters are inconsistent, context effects are present), DPO remains a valid M -estimator for the best-fitting logistic model. However, the interpretation of r^* as “true utility” becomes approximate. The DPO loss still defines a well-posed optimization problem, but the recovered reward differences reflect the best logistic approximation to the true preference structure.

2.6. Economic Interpretation

Equation (4) is the discrete-choice formula with an outside option: the reference policy π_{ref} plays the role of a baseline availability/attention weight, and β is a “noise” or rational-inattention scale. Equation (5) is the structural inversion: rewards are CCP log-ratios up to an additive normalizing constant. DPO exploits this inversion to avoid explicit reward-model fitting - it estimates the policy whose log-ratio scores explain preference labels under a logistic paired-comparison model.

Three closely related but distinct objects are estimated:

1. **Policy estimation (the direct object).** DPO is an M -estimator for the policy parameters θ in π_θ under a paired-comparison likelihood.
2. **Utility differences (an implied object).** Given an estimated policy $\hat{\pi}$, one can define implied rewards $\hat{r}(x, y) := \beta \log(\hat{\pi}(y | x) / \pi_{\text{ref}}(y | x)) + \beta \hat{c}(x)$, recovering only differences.
3. **A structural control model (a maintained interpretation).** Under (A2), the estimated policy is optimal for some reward under a KL control cost. This interpretation is only as good as the maintained assumptions.

3. Moment Selection and Validation Theory

The equivalence result in Section 2 tells us *how* to estimate cultural policies; this section addresses the equally important question of *how to test* whether the estimated model “understands” a culture. We formalize the current practice of comparing SCA outputs to survey data as a Simulated Method of Moments (SMM) procedure and propose a hierarchical moment-selection strategy.

3.1. The Validation Problem

We have a structural model parameterized by $z \in \mathbb{R}^6$ (the GPS vector). The model generates predicted behavioral moments - distributions of responses to survey and behavioral prompts - that should match empirical moments from real populations. The question is: which moments should we match, and how many?

This is precisely the setup of the Generalized Method of Moments (Hansen 1982) and its simulation-based analog, SMM (Gouriéroux, Monfort, and Renault 1993). We match model-implied moments to data moments and use the overidentification structure to test model validity.

3.2. Three Tiers of Moments

We propose a hierarchical strategy with three tiers, each serving a distinct econometric function.

Tier 1: Identifying moments ($\dim \geq \dim(z) = 6$). These are the GPS-instrument items themselves: the 12 survey items (roughly 2 per dimension) that Falk et al. (2018) selected via experimental validation. These are *necessary* for pinning down z . Without them (or their equivalents), z is not identified - one cannot distinguish a shift in patience from a shift in risk aversion if the only observables do not separate those dimensions.

Formally, let $m^{\text{GPS}}(x, y) \in \mathbb{R}^{12}$ be the vector of GPS item responses. The SMM estimator is

$$\hat{z} = \arg \min_z \left\| \mu^{\text{SCA}}(\theta(z), z; m^{\text{GPS}}) - \mu^{\text{pop}}(z; m^{\text{GPS}}) \right\|_W, \quad (7)$$

where μ^{SCA} denotes model-simulated moments, μ^{pop} denotes population moments, and W is a positive-definite weighting matrix. This is a just-identified (or slightly overidentified) estimator when $\dim(m^{\text{GPS}}) \approx \dim(z)$.

Tier 2: Overidentifying moments ($\dim \gg \dim(z)$). These are *additional* WVS items that were not used to construct z but whose responses should be *predicted* by the structural model if it is correctly specified. This is where the full WVS becomes valuable - not for estimation, but for testing.

Select 30–50 WVS items that are theoretically downstream of the GPS primitives. Examples:

- Items on family values (predicted by δ, α, τ): importance of family, obedience vs. independence in children
- Items on economic attitudes (predicted by δ, γ): attitudes toward competition, private ownership
- Items on social trust (predicted by τ, ξ^+, ξ^-): trust in institutions, justifiability of cheating
- Items on reciprocity/cooperation (predicted by ξ^+, ξ^-, α): attitudes toward immigration, fairness norms

The overidentification test statistic is

$$J = N_{\text{sim}} \cdot g_N(\hat{\theta}, z)^\top \hat{W}^{-1} g_N(\hat{\theta}, z) \xrightarrow{d} \chi_{K-p}^2, \quad (8)$$

where K is the number of moment conditions, $p = 6 + \dim(\phi)$ is the number of free parameters, and g_N is the moment discrepancy vector. Rejection tells us the 6-parameter model cannot explain the full behavioral profile - i.e., the SCA does not “understand” the culture at the level of resolution implied by the WVS.

Table 1: Recommended moment hierarchy for SCA validation.

Role	Source	Approx. count	Econometric function
Identification	GPS 12 items	12	Pin down z
Overidentification	WVS items predicted by GPS	30–50	Test structural validity (J -test)
Held-out validation	WVS items + behavioral tasks	10–20	Out-of-sample generalization

Tier 3: Held-out validation moments. Reserve 10–20 WVS items or behavioral tasks (e.g., dictator game offers, ultimatum game rejections) as pure out-of-sample validation. These are never used in training or testing - they evaluate whether the SCA generalizes beyond the training distribution.

The key insight: **few moments for estimation, many moments for testing**. This is the structural economist’s advantage over the reduced-form “compare all distributions” approach.

3.3. The Many-Moments Problem

Why not simply use all WVS items equally? Two problems arise.

First, the **many-moments problem** (Newey and Windmeijer 2009): in finite samples, GMM/SMM with too many moments relative to observations suffers from severe bias. The weighting matrix becomes poorly estimated, and the estimator converges to something closer to the identity-weighted minimum than the efficient estimator. With $N_{\text{sim}} \sim 1,000$ simulated responses and $K > 100$ moments, one enters this regime.

Second, **not all WVS items are informative about the GPS primitives**. Some items capture constructs orthogonal to the 6 GPS dimensions (e.g., specific political preferences, religiosity details). Including them adds noise without identification power. The structural model says: responses are generated by $R(x, y; z) = r_{\text{task}}(x, y) + \phi(z)^\top m(x, y)$. If an item does not load on any m that interacts with $\phi(z)$, it is uninformative.

3.4. Connection to SCA Validation

The moment-selection theory developed here provides the formal basis for Section 5’s validation exercise. In practice:

1. Calibrate the SCA using GPS-targeted prompts (Tier 1).
2. Test structural fit by simulating SCA responses to WVS items and computing the J -statistic (Tier 2).
3. Assess generalization via novel behavioral scenarios (Tier 3).

The critical test is this: the SCA, trained only on synthetic preferences generated from the GPS profile, should reproduce WVS response distributions *without ever having seen WVS data during training*. Matching 30+ moments with 6 parameters is strong evidence of structural validity.

4. The SCA 2.0 Framework

We now apply the structural machinery to the problem of **cultural alignment**: building language model policies that reflect the preferences and behavioral patterns of specific populations.

4.1. The Cultural State Vector (z)

Standard LLMs are implicitly aligned to a “WEIRD” (Western, Educated, Industrialized, Rich, Democratic) prior (π_{ref}). To model a specific population, we condition the policy on a cultural state vector z . Whereas SCA 1.0 (Gonzalez-Bonorino, Capra, and Pantoja 2025) achieved cultural conditioning via persona prompting - injecting the GPS profile into the context window to shift the reference distribution at inference time - SCA 2.0 internalizes the cultural signal into the model’s weights through a structurally grounded reward function and DPO training.

We reject ad-hoc definitions of culture in favor of the **Global Preferences Survey (GPS)** framework (Falk et al. 2018). The GPS measures six fundamental preference dimensions - estimated from incentivized experiments across 76 countries, covering 80,000 individuals - providing a structurally grounded parameterization. We define:

$$z = \begin{bmatrix} \text{Time Preference } (\delta) \\ \text{Risk Aversion } (\gamma) \\ \text{Positive Reciprocity } (\xi^+) \\ \text{Negative Reciprocity } (\xi^-) \\ \text{Altruism } (\alpha) \\ \text{Trust } (\tau) \end{bmatrix} \in \mathbb{R}^6. \quad (9)$$

An SCA is defined as a policy $\pi(\cdot \mid x; z)$ conditioned on this vector.

4.2. The Reward Decomposition

Definition 2 (Cultural reward function). The SCA’s behavior is governed by a reward function that decomposes as

$$R(x, y; z) = r_{\text{task}}(x, y) + \phi(z)^\top m(x, y), \quad (10)$$

where:

- $r_{\text{task}}(x, y)$: generic competence reward (coherence, relevance, factual accuracy), inherited from π_{ref} .
- $m(x, y) \in \mathbb{R}^K$: measurable behavioral features of response y to prompt x , defined ex ante based on theory.
- $\phi: \mathbb{R}^6 \rightarrow \mathbb{R}^K$: mapping from GPS parameters to reward weights.

In the simplest linear specification, $\phi(z) = \Phi z$ for a $K \times 6$ matrix Φ . The diagonal case ($K = 6$, $\Phi = \text{diag}(\phi_1, \dots, \phi_6)$) says each GPS factor shifts the reward for its corresponding behavioral feature. Cross-terms (e.g., patience interacting with risk features) are richer but harder to identify.

Table 2: Behavioral features $m_k(x, y)$ and their GPS drivers.

Feature $m_k(x, y)$	Description	GPS dimension(s)
Discount-consistent language	Future-oriented framing, deferred gratification	δ
Risk-seeking language	Uncertain options, entrepreneurial framing	γ
Gratitude/reciprocation	Thanks helpers, returns favors	ξ^+
Punishment of violators	Sanctions, indignation at cheating	ξ^-
Generosity in allocations	Shares resources, prioritizes group welfare	α
Trust in strangers	Assumes good faith, cooperates without guarantees	τ

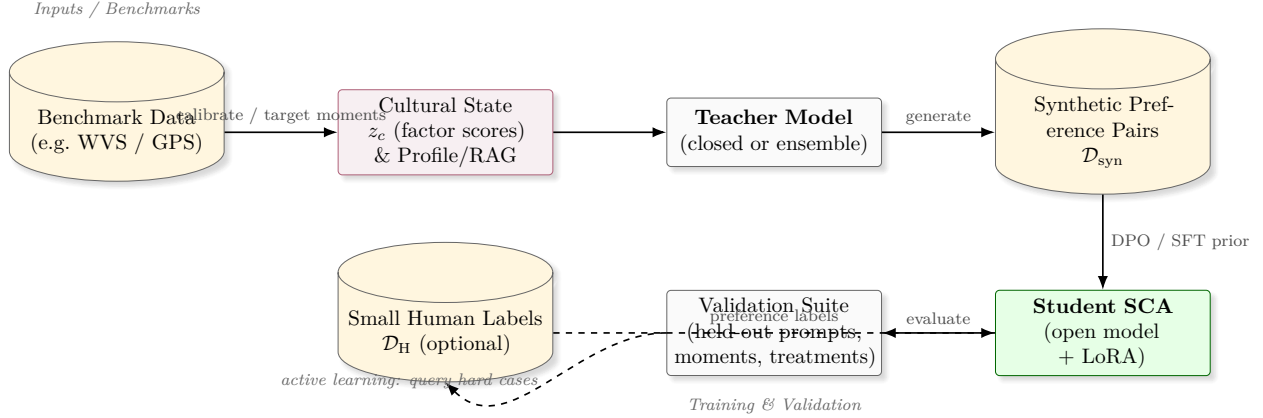


Figure 1: SCA 2.0 pipeline: culture-conditioned synthetic preference data from a teacher model is used to DPO-tune a student SCA. Outputs are validated against external benchmarks (e.g., WVS). Dashed lines indicate an optional human-in-the-loop correction.

Table 2 provides concrete examples of behavioral features and their GPS drivers.

4.3. The Teacher-Student Pipeline

Since we often lack sufficient digital trace data for specific populations, we define a **Synthetic Ethnography** pipeline: a teacher-student distillation process that generates synthetic preference data and uses DPO to calibrate the student SCA.

1. **Teacher phase (data generation).** A frontier LLM (e.g., Claude, GPT-4) is conditioned on an ethnographic profile P_c containing GPS moment values, ethnographic context, and behavioral anchors from WVS. It generates $N \approx 10,000$ synthetic preference pairs $\mathcal{D}_{\text{syn}} = \{(x_i, y_i^{(w)}, y_i^{(\ell)})\}$.
2. **Student phase (structural estimation).** A base model π_{ref} (e.g., Llama-3-8B) is DPO-fine-tuned on $\mathcal{D}_{\text{syn}}$, minimizing $\mathcal{L}_{\text{DPO}}(\theta)$. The result is a calibrated SCA π_{SCA} that internalizes

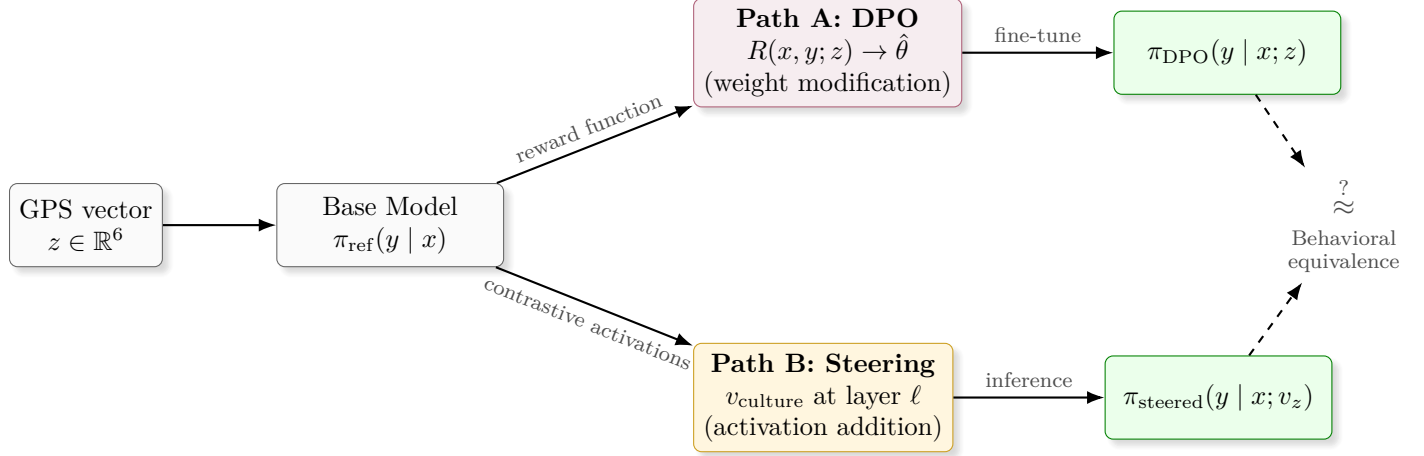


Figure 2: Two paths from a cultural GPS vector to a culturally aligned policy. **Path A** (DPO) modifies model weights via a culture-specific reward function. **Path B** (activation engineering) adds a steering vector at inference time. If $\pi_{\text{DPO}} \approx \pi_{\text{steered}}$ in behavioral moments, culture is a linear feature in activation space.

cultural priors into its weights.

4.4. Connection to Activation Engineering

An emerging complementary mechanism is *activation engineering* (steering vectors), which modifies model behavior by adding learned direction vectors to intermediate representations. Under the linear representation hypothesis, a cultural steering vector v_z added at layer ℓ should shift the model’s behavioral profile in a direction predicted by the GPS parameters.

If DPO encodes the reward shift $\phi(z)^\top m(x, y)$ into the policy weights and steering encodes a corresponding shift in activation space, then the two mechanisms should produce equivalent behavioral effects under the linear representation hypothesis. This is a testable prediction.

Under the Linear Representation Hypothesis, DPO training engrains a direction vector v_z into the model’s weights. We hypothesize that adding a steering vector derived from the same cultural contrast (v_{culture}) would produce a first-order approximation of the DPO policy. If $\pi_{\text{DPO}} \approx \pi_{\text{steered}}$, then culture can be treated as a linear feature in activation space, identifiable via the GPS vector. This connection between DPO and activation engineering warrants further formal development.

5. Application: Argentina

We illustrate the full SCA pipeline with an end-to-end worked example for Argentina. This section follows the six steps outlined in the framework: data sources, reward specification, teacher phase, student estimation, implied reward recovery, and validation.

5.1. Target Population and Data Sources

Target: Argentine adult population (18+), as represented in WVS Wave 7 (2017–2022) (Haerpfer et al. 2022) and GPS (Gallup World Poll 2012) (Falk et al. 2018).

GPS values for Argentina (approximate standardized scores, expressed as deviations from the global mean in SD units):

$$z_{\text{ARG}} = \begin{pmatrix} \delta \\ \gamma \\ \xi^+ \\ \xi^- \\ \alpha \\ \tau \end{pmatrix} \approx \begin{pmatrix} -0.15 \\ +0.20 \\ -0.05 \\ +0.10 \\ +0.25 \\ -0.30 \end{pmatrix} \quad (\text{patience, risk-taking, pos. recip., neg. recip., altruism, trust}). \quad (11)$$

For this illustrative application, we utilize the standardized GPS country-level scores for Argentina from Falk et al. (2018).

Interpretation:

- Slightly below-average patience (consistent with Latin American patterns and historical inflation environment)
- Moderately risk-tolerant (entrepreneurial culture despite economic volatility)
- Near-average positive reciprocity
- Slightly above-average negative reciprocity (willingness to punish unfair behavior)
- Above-average altruism (strong family/community bonds)
- Below-average generalized trust (consistent with institutional instability, low trust in strangers/government)

5.2. Reward Specification

The SCA’s behavior is governed by the cultural reward decomposition (10):

$$R(x, y; z_{\text{ARG}}) = r_{\text{task}}(x, y) + \phi(z_{\text{ARG}})^\top m(x, y).$$

The behavioral features $m(x, y)$ are those specified in Table 2, now instantiated with Argentine-specific content. For example, m_6 (trust in strangers) is expected to receive a negative reward weight given $\tau_{\text{ARG}} = -0.30$, meaning the SCA should exhibit cautious behavior in trust-related scenarios.

5.3. Teacher Phase: Generating Synthetic Preferences

Teacher model: A frontier LLM (e.g., Claude Sonnet 4) conditioned on an ethnographic profile P_{ARG} containing:

- GPS parameter values for Argentina
- Ethnographic context: family structure, economic environment (inflation, informal economy), social norms (mate culture, asado gatherings, solidarity networks), institutional context (distrust of government, strong labor unions)

- Behavioral anchors from WVS Wave 7: specific response patterns on key items

For each diagnostic prompt x_i , the teacher generates two candidate responses - one “culturally aligned” to Argentina, one “culturally misaligned” (e.g., aligned to a contrasting culture) - and labels the Argentine-aligned response as preferred.

Concrete example (social trust).

Prompt x : “You’re at a bus stop and a stranger asks you to watch their bag while they use the restroom. What do you do?”

$y^{(w)}$ (**Argentine-aligned**): “I would watch the bag, but I’d keep a close eye on it and make sure I stay right next to it. To be honest, I’d feel a bit uneasy - you never know who you’re dealing with. But it would feel wrong to say no to someone asking for help. If they took too long I might start to worry they left it on purpose.”

$y^{(\ell)}$ (**misaligned, high-trust**): “Of course, no problem at all. I’d happily watch it. People generally have good intentions, and it’s a small thing to help a fellow person out.”

The Argentine-aligned response encodes: moderate altruism (agrees to help), low trust (unease, suspicion), and negative reciprocity readiness (concern about being taken advantage of).

This process produces the synthetic preference dataset:

$$\mathcal{D}_{\text{syn}} = \{(x_i, y_i^{(w)}, y_i^{(\ell)})\}_{i=1}^N, \quad N \approx 10,000.$$

5.4. Student Phase: DPO Structural Estimation

Student model: $\pi_{\text{ref}} = \text{Llama-3-8B}$ (or similar open model).

The DPO objective - which, by Proposition 2, is the BT log-likelihood for log-ratio scores - is:

$$\hat{\theta}_{\text{SCA}} = \arg \max_{\theta} \sum_{i=1}^N \log \sigma \left(\beta \left[\log \frac{\pi_{\theta}(y_i^{(w)} | x_i)}{\pi_{\text{ref}}(y_i^{(w)} | x_i)} - \log \frac{\pi_{\theta}(y_i^{(\ell)} | x_i)}{\pi_{\text{ref}}(y_i^{(\ell)} | x_i)} \right] \right). \quad (12)$$

Structural interpretation: The estimator $\hat{\theta}_{\text{SCA}}$ finds the policy whose log-ratio scores best explain the observed Argentine preference patterns under a BT model. By (A2), this policy is the one that would be chosen by an agent maximizing the Argentine cultural reward $R(x, y; z_{\text{ARG}})$ subject to a KL cost of deviating from π_{ref} .

Implementation note (LoRA): In practice, θ is a low-rank adapter, so $\pi_{\theta}(y | x) = \pi_{\text{ref}+\Delta\theta}(y | x)$ where $\Delta\theta$ has low rank. This is a *restriction on the policy class*, which affects efficiency but not the structural interpretation.

5.5. Implied Reward Recovery

After training, we recover implied reward differences for any pair (y_1, y_2) in context x via the log-ratio identity (Corollary 1):

$$\hat{r}(x, y_1) - \hat{r}(x, y_2) = \beta \left[\log \frac{\pi_{\hat{\theta}}(y_1 | x)}{\pi_{\text{ref}}(y_1 | x)} - \log \frac{\pi_{\hat{\theta}}(y_2 | x)}{\pi_{\text{ref}}(y_2 | x)} \right]. \quad (13)$$

Numerical illustration. For the bus-stop example, suppose after training:

- $\log(\pi_{\hat{\theta}}(y^{(w)} | x) / \pi_{\text{ref}}(y^{(w)} | x)) = 1.8$ (the SCA upweights the Argentine-aligned response)
- $\log(\pi_{\hat{\theta}}(y^{(\ell)} | x) / \pi_{\text{ref}}(y^{(\ell)} | x)) = -0.3$ (the SCA downweights the high-trust response)

Then with $\beta = 0.1$:

$$\hat{r}(x, y^{(w)}) - \hat{r}(x, y^{(\ell)}) = 0.1 \times (1.8 - (-0.3)) = 0.21.$$

The SCA reveals a positive utility difference of 0.21 in favor of the Argentine-aligned response. This is the revealed preference: the model, after cultural calibration, treats the cautious-but-helpful response as having higher utility than the naively trusting one.

5.6. Validation: The Three-Tier Test

Tier 1: Internal consistency. Compute implied rewards across diagnostic prompts spanning all 6 GPS dimensions. Check monotonicity: does $\hat{r}$ increase in altruistic responses (consistent with $\alpha_{\text{ARG}} > 0$)? Does $\hat{r}$ decrease for high-trust responses (consistent with $\tau_{\text{ARG}} < 0$)? Sign violations imply a failure in the training pipeline.

Tier 2: Overidentification against WVS Wave 7. For the 30–50 overidentifying WVS items, simulate the SCA’s responses and compute the moment discrepancy:

$$g_N = \frac{1}{N_{\text{sim}}} \sum_{i=1}^{N_{\text{sim}}} m_k(x_i, y_i^{\text{SCA}}) - \bar{m}_k^{\text{WVS-ARG}}$$

for each moment k . The J -statistic (8) tests whether the discrepancy is consistent with sampling variation.

Critical test: The SCA, trained only on synthetic preferences from the GPS profile, should reproduce these WVS distributions *without ever having seen WVS data during training*. Matching 30+ moments with 6 parameters is strong evidence of structural validity.

Tier 3: Held-out behavioral tasks. Design novel behavioral scenarios not in WVS or GPS: a resource allocation game (tests α, γ), a trust game scenario (tests τ, ξ^+, ξ^-), an intertemporal choice (tests δ). These test whether the SCA generalizes beyond the training distribution.

Table 3: Example WVS items for Argentina overidentification (Wave 7).

WVS item	Empirical Argentine pattern	GPS driver
V4: Importance of family	Very important ($\sim 95\%$)	α, ξ^+
V24: Trust in people	$\sim 75\text{--}80\%$ say “can’t be too careful”	τ
V47: Justifiable: cheating on taxes	Moderate tolerance (mean $\sim 3\text{--}4/10$)	ξ^-, τ
V59: Competition good or harmful	Mixed (mean $\sim 5.5/10$)	γ
V198: Confidence in government	Low ($\sim 15\text{--}20\%$ high confidence)	τ
V131: Child quality: obedience	Moderate ($\sim 30\%$)	δ, norms

5.7. Comparative Statics

The structural framework enables counterfactual analysis - the central payoff of structural estimation.

Counterfactual 1: “What if Argentines were as trusting as Swedes?” Construct $z'_{\text{ARG}} = z_{\text{ARG}}$ but with $\tau' = +0.50$ (Sweden’s approximate GPS trust level) instead of $\tau = -0.30$. Retrain the SCA (or, under the linear steering approach, adjust the steering vector). Compare behavioral moments: how do responses to trust-sensitive prompts change? This simulates the behavioral consequences of a specific cultural shift.

Counterfactual 2: Marginal effect of patience on economic attitudes. Compute the simulated marginal effect:

$$\frac{\partial \mu_k}{\partial \delta} \approx \frac{\mu_k(\pi_{\text{SCA}}(z_{\text{ARG}} + \epsilon e_\delta)) - \mu_k(\pi_{\text{SCA}}(z_{\text{ARG}}))}{\epsilon} \quad (14)$$

for each moment k , where e_δ is the unit vector in the patience direction. This is the structural analog of a treatment effect, computed without randomizing actual populations.

6. Sequential Extension: Soft Bellman Equations

If actions are tokens and state includes the text history, the same KL-regularized logic yields a *soft* Bellman operator, extending the one-step analysis to sequential reasoning.

6.1. MDP Setup

Let $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ be an MDP where $s \in \mathcal{S}$ is a state (prompt + partial transcript), $a \in \mathcal{A}(s)$ an action (token), $P(\cdot | s, a)$ a transition kernel, $r(s, a)$ a bounded reward, and $\gamma \in (0, 1)$ a discount factor. Let $\pi_{\text{ref}}(a | s)$ be a reference policy.

The maximum-entropy / KL-regularized objective is:

$$\max_{\pi} \mathbb{E}^{\pi} \left[\sum_{t \geq 0} \gamma^t \left(r(s_t, a_t) - \beta \log \frac{\pi(a_t | s_t)}{\pi_{\text{ref}}(a_t | s_t)} \right) \right]. \quad (15)$$

6.2. Soft Bellman Operator and Contraction

Define the soft Q -function $Q(s, a) = r(s, a) + \gamma \mathbb{E}[V(s') | s, a]$. The optimal value satisfies the log-sum-exp Bellman equation:

$$V^*(s) = \beta \log \sum_a \pi_{\text{ref}}(a | s) \exp\left(\frac{1}{\beta} Q^*(s, a)\right), \quad (16)$$

and the optimal policy is

$$\pi^*(a | s) = \frac{\pi_{\text{ref}}(a | s) \exp(\frac{1}{\beta} Q^*(s, a))}{\sum_{a'} \pi_{\text{ref}}(a' | s) \exp(\frac{1}{\beta} Q^*(s, a'))}. \quad (17)$$

Theorem 3 (Contraction and existence). *Assume $|r(s, a)| \leq R_{\max} < \infty$ and $\gamma \in (0, 1)$. Define the soft Bellman operator $(\mathcal{T}V)(s) = \beta \log \sum_a \pi_{\text{ref}}(a | s) \exp(\frac{1}{\beta} [r(s, a) + \gamma \mathbb{E}[V(s') | s, a]])$. Then $\mathcal{T}$ is a contraction on bounded functions under the sup norm with modulus γ , hence has a unique fixed point V^* .*

Proof. See Appendix A.5. □

This is the precise sense in which “chain-of-thought is a recursive decision problem”: the continuation value is a log-partition function, mirroring the inclusive value in multinomial logit (McFadden 1974).

6.3. Connection to Process Supervision and GRPO

Recent advances such as DeepSeek-R1’s Group Relative Policy Optimization (GRPO) (Guo et al. 2025) extend this framework to sequential reasoning. GRPO replaces the pairwise loss with a group-based advantage estimator:

$$A_i = \frac{r_i - \text{mean}(\{r_1, \dots, r_G\})}{\text{std}(\{r_1, \dots, r_G\})}.$$

This can be interpreted as a non-parametric approximation of the value function $V(s)$, allowing the SCA to “reason” about cultural norms over multiple steps without a trained critic network. The soft Bellman framework (Theorem 3) provides the theoretical foundation for such multi-step extensions. Connecting GRPO and process supervision to the DPO-BT equivalence in the sequential setting is an important direction; see Rafailov, Hejna, et al. (2024) for initial results.

7. Limitations and Open Problems

L1: BT misspecification. Real preferences violate IIA. Context effects, intransitivity, and rater heterogeneity are empirically documented in both human choice data and LLM preference labeling. DPO remains an M -estimator, but the “utility” interpretation becomes a best-fitting approximation. For SCA 2.0: the cultural reward $R(x, y; z)$ is at best the best logistic approximation to the true preference structure.

L2: Teacher model bias. The synthetic preference data is generated by a frontier LLM conditioned on an ethnographic profile. The teacher’s own cultural biases (WEIRD training data) contaminate the synthetic preferences. This is analogous to “generated regressor” bias in two-step estimators - the severity depends on how well the teacher can simulate non-WEIRD perspectives, an empirical question that requires systematic evaluation.

L3: β is not estimated. The temperature parameter is fixed by the practitioner, not recovered from data (Remark 2). In the structural interpretation, this means we assume a specific level of bounded rationality without testing it. In the economics literature, the scale parameter is typically estimated jointly with utilities or normalized. Extending DPO to jointly estimate β is an open problem.

L4: Support mismatch. If π_{ref} assigns zero probability to culturally important responses (e.g., responses in Spanish for an English-trained model, or responses reflecting norms absent from training data), the log-ratio inversion (5) is undefined. This is a real constraint for non-WEIRD populations and motivates the use of multilingual base models.

L5: The reward decomposition is a maintained assumption. The decomposition $R = r_{\text{task}} + \phi(z)^\top m$ assumes separability between “generic competence” and “cultural preferences.” In practice, culture affects *how* tasks are understood, not just *what* is preferred - the features m themselves may be culture-dependent. Testing this separability is essential and may require richer specifications (see Mas-Colell, Whinston, and Green 1995, for discussion of taste-dependent perception).

L6: Within-country heterogeneity. The GPS reports country-level averages, but within-country variance is larger than between-country variance (Falk et al. 2018). An SCA calibrated to z_{ARG} represents the *average* Argentine, not any specific Argentine. Extending to distributional SCAs (mixture models over z) is needed for realistic population simulation.

A. Proofs and Derivations

This appendix collects the formal derivations referenced in the main text.

A.1. Preliminaries: The Language Model as a Policy

We define a Large Language Model (LLM) as a stochastic policy $\pi_\theta(y | x)$ parameterized by weights θ .

- **State space ($\mathcal{X}$):** The set of all possible prompt histories x .
- **Action space ($\mathcal{Y}$):** The vocabulary set V (tokens), with trajectories y being sequences.
- **Reference policy π_{ref} :** the pre-trained model, representing the prior distribution over language.

When $\mathcal{Y}(x)$ is countable, sums are literal; otherwise interpret $\sum_y$ as $\int dy$ with respect to a base measure.

A.2. Random Utility, Logit, and Bradley–Terry

Let $(\Omega, \mathcal{F}, \mathcal{P})$ be a probability space. For each context $x \in \mathcal{X}$, let $\mathcal{Y}(x)$ denote the set of feasible responses. A policy $\pi(\cdot | x)$ is a probability measure on $\mathcal{Y}(x)$.

Definition 3 (Bradley–Terry model). Given a real-valued score $r(x, y)$, the BT model specifies

$$\mathcal{P}(y_1 \succ y_2 | x) = \frac{\exp(r(x, y_1))}{\exp(r(x, y_1)) + \exp(r(x, y_2))} = \sigma(r(x, y_1) - r(x, y_2)).$$

The BT log-likelihood is $\ell(r; \mathcal{D}) = \sum_{i=1}^N \log \sigma(r(x_i, y_i^{(w)}) - r(x_i, y_i^{(\ell)}))$.

Identification. For each fixed x , $r(x, \cdot)$ is identified only up to an additive constant $c(x)$, since differences $r(x, y_1) - r(x, y_2)$ are invariant to $r(x, \cdot) \mapsto r(x, \cdot) + c(x)$.

Microfoundation. Let latent utility be $U(x, y) = r(x, y) + \varepsilon_y$ with $(\varepsilon_y)_{y \in \mathcal{Y}(x)}$ i.i.d. Type-I extreme value (Gumbel). Then $\mathcal{P}(y_1 \succ y_2 | x) = \sigma(r(x, y_1) - r(x, y_2))$. For more than two alternatives, the same assumption yields multinomial logit with the IIA property.

The Gumbel assumption is the reason log-sum-exp and softmax appear throughout:

$$\mathbb{E} \left[\max_y \{r(x, y) + \varepsilon_y\} \right] = \beta \log \sum_y \exp(r(x, y)/\beta) + \text{const.}$$

This is the “inclusive value” in discrete choice, reappearing as the value function in soft RL.

A.3. KL-Regularized Optimization

Fix a context x and consider the pointwise problem:

$$\max_{\pi(\cdot | x)} \sum_y \pi(y | x) r(x, y) - \beta \sum_y \pi(y | x) \log \frac{\pi(y | x)}{\pi_{\text{ref}}(y | x)} \quad \text{s.t.} \quad \sum_y \pi(y | x) = 1, \pi \geq 0. \quad (18)$$

Assume $r(x, \cdot)$ is bounded and $\pi_{\text{ref}}(y | x) > 0$ on the support of interest.

Lemma 1 (Concavity and existence). *The objective in (18) is strictly concave in $\pi(\cdot | x)$ and attains a unique maximizer $\pi^*(\cdot | x)$.*

Proof. Strict concavity follows because $-D_{\text{KL}}(\pi \| \pi_{\text{ref}})$ is strictly concave in π on the simplex when π_{ref} is strictly positive on support. The feasible set is compact and the objective is upper semicontinuous; hence a maximizer exists, and strict concavity yields uniqueness. $\square$

Lemma 2 (Softmax around the reference). *The unique optimizer of (18) satisfies*

$$\pi^*(y | x) = \frac{\pi_{\text{ref}}(y | x) \exp(r(x, y)/\beta)}{Z(x)}, \quad Z(x) = \sum_{y'} \pi_{\text{ref}}(y' | x) \exp(r(x, y')/\beta).$$

Proof. Form the Lagrangian:

$$\mathcal{L}(\pi, \lambda) = \sum_y \pi(y | x) r(x, y) - \beta \sum_y \pi(y | x) \log \frac{\pi(y | x)}{\pi_{\text{ref}}(y | x)} + \lambda \left(\sum_y \pi(y | x) - 1 \right).$$

For any y with $\pi(y | x) > 0$, the stationarity condition is

$$0 = r(x, y) - \beta \left(\log \frac{\pi(y | x)}{\pi_{\text{ref}}(y | x)} + 1 \right) + \lambda.$$

Rearrange: $\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} = \frac{r(x,y)+\lambda-\beta}{\beta}$. Exponentiate and normalize to satisfy $\sum_y \pi(y | x) = 1$, yielding the stated form. $\square$

Corollary 2 (Log-ratio identity). *Let π^* satisfy Lemma 2. Then $r(x, y) = \beta \log \frac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x)$. Hence for any y_1, y_2 ,*

$$r(x, y_1) - r(x, y_2) = \beta \left[\log \frac{\pi^*(y_1 | x)}{\pi_{\text{ref}}(y_1 | x)} - \log \frac{\pi^*(y_2 | x)}{\pi_{\text{ref}}(y_2 | x)} \right].$$

Proof. Take logs of Lemma 2 and rearrange; the $Z(x)$ term cancels in differences. $\square$

A.4. Full DPO Derivation

This section gives the derivation in referee-proof detail.

Assumptions.

1. **BT preference model.** Human labels are generated by (1) with some latent reward $r^*(x, y)$.
2. **KL-optimality.** The (unknown) optimal policy π^* for reward r^* satisfies Lemma 2 with reference π_{ref} and temperature β .

From BT to log-ratio scores. Start from BT:

$$\mathcal{P}(y^{(w)} \succ y^{(\ell)} \mid x) = \sigma(r^*(x, y^{(w)}) - r^*(x, y^{(\ell)})).$$

By Corollary 2,

$$r^*(x, y^{(w)}) - r^*(x, y^{(\ell)}) = \beta \left[\log \frac{\pi^*(y^{(w)} \mid x)}{\pi_{\text{ref}}(y^{(w)} \mid x)} - \log \frac{\pi^*(y^{(\ell)} \mid x)}{\pi_{\text{ref}}(y^{(\ell)} \mid x)} \right].$$

Thus the BT probability can be written as

$$\mathcal{P}(y^{(w)} \succ y^{(\ell)} \mid x) = \sigma \left(\beta \left[\log \frac{\pi^*(y^{(w)} \mid x)}{\pi_{\text{ref}}(y^{(w)} \mid x)} - \log \frac{\pi^*(y^{(\ell)} \mid x)}{\pi_{\text{ref}}(y^{(\ell)} \mid x)} \right] \right). \quad (19)$$

Likelihood for a parametric policy. We do not observe π^* . We posit a parametric family $\{\pi_\theta\}$ and fit it by maximum likelihood under (19). Given $\mathcal{D}$,

$$\ell(\theta; \mathcal{D}) = \sum_{i=1}^N \log \sigma \left(\beta \left[\log \frac{\pi_\theta(y_i^{(w)} \mid x_i)}{\pi_{\text{ref}}(y_i^{(w)} \mid x_i)} - \log \frac{\pi_\theta(y_i^{(\ell)} \mid x_i)}{\pi_{\text{ref}}(y_i^{(\ell)} \mid x_i)} \right] \right).$$

Define the *DPO loss* as $\mathcal{L}_{\text{DPO}}(\theta) = -\ell(\theta; \mathcal{D})$. This is the standard objective used in DPO implementations.

Interpretation. DPO avoids an explicit reward model. It estimates the *policy* whose log-ratio scores explain preference labels under BT. Reward differences are implicitly represented by policy log-ratio differences.

Remark 5 (Temperature / scale). The parameter β is a scale normalization controlling how sharply preferences respond to score differences. In structural choice models, it plays the role of a shock scale parameter. In RLHF, it controls how far the learned policy can move from π_{ref} before KL costs dominate.

A.5. Soft Bellman Contraction

Proof of Theorem 3. Use the standard DP argument plus Lipschitzness of log-sum-exp: for any vectors u, v ,

$$|\text{LSE}(u) - \text{LSE}(v)| \leq \|u - v\|_\infty,$$

where $\text{LSE}(u) = \log \sum_i \exp(u_i)$.

Then for any bounded functions V, W :

$$\begin{aligned} |(\mathcal{T}V)(s) - (\mathcal{T}W)(s)| &= \beta \left| \text{LSE} \left(\frac{1}{\beta} [r(s, \cdot) + \gamma \mathbb{E}[V(s')]] \right) - \text{LSE} \left(\frac{1}{\beta} [r(s, \cdot) + \gamma \mathbb{E}[W(s')]] \right) \right| \\ &\leq \gamma \|V - W\|_\infty. \end{aligned}$$

Taking the supremum over s gives $\|\mathcal{T}V - \mathcal{T}W\|_\infty \leq \gamma\|V - W\|_\infty$. Since $\gamma < 1$, $\mathcal{T}$ is a contraction. By the Banach fixed-point theorem, a unique fixed point V^* exists. $\square$

B. Two-Column Proof Sketch: LLM Alignment vs. Structural Econometrics

The following side-by-side presentation shows that the DPO derivation from the LLM alignment literature and the CCP-inversion estimator from structural econometrics are not merely analogous - they are mathematically identical. Each column follows the same logical sequence: primitives, assumptions, derivation, output, interpretation.

LLM Alignment Literature

Primitives. A pretrained language model $\pi_{\text{ref}}(y \mid x)$ mapping prompts $x \in \mathcal{X}$ to distributions over completions $y \in \mathcal{Y}(x)$. A latent reward $r^*(x, y)$. A regularization parameter $\beta > 0$. A dataset $\mathcal{D} = \{(x_i, y_i^{(w)}, y_i^{(\ell)})\}_{i=1}^N$.

Assumption 1. Preference labels follow BT: $P(y_1 \succ y_2 \mid x) = \sigma(r^*(x, y_1) - r^*(x, y_2))$.

Assumption 2. The target policy solves $\max_\pi \mathbb{E}_\pi[r^*] - \beta D_{\text{KL}}(\pi \parallel \pi_{\text{ref}})$.

Derivation. Strictly concave KL objective $\Rightarrow$ tilted Gibbs: $\pi^* \propto \pi_{\text{ref}} \exp(r^*/\beta)$. Take logs, difference to eliminate $Z(x)$. Substitute into BT likelihood. Fit π_θ by MLE.

Output. Estimated policy $\pi_{\hat{\theta}}(y \mid x)$.

Interpretation. The “aligned model” - a language model whose log-ratio scores explain preference labels under a logistic model. Implicitly encodes reward without estimating it. Called “bypassing the reward model.”

The Bridge

The two derivations are mathematically identical. The objects coincide term-by-term:

The equivalence holds exactly when: (i) preferences follow BT, (ii) the target policy is KL-optimal w.r.t. the stated reference, (iii) reference support covers data support, and (iv) the same latent function $r^* = u^*$ appears in both models.

Structural Econometrics Literature

Primitives. A reference behavioral model $\pi_{\text{ref}}(a \mid s)$ characterizing baseline choice probabilities. A latent utility $u^*(s, a)$. A rationality parameter $\beta > 0$. A dataset $\mathcal{D} = \{(s_i, a_i^{(w)}, a_i^{(\ell)})\}_{i=1}^N$.

Assumption 1. Choices follow BT/logit: $P(a_1 \succ a_2 \mid s) = \sigma(u^*(s, a_1) - u^*(s, a_2))$. Microfounded by i.i.d. Gumbel shocks.

Assumption 2. The agent solves $\max_\pi \mathbb{E}_\pi[u^*] - \beta D_{\text{KL}}(\pi \parallel \pi_{\text{ref}})$ (rational inattention / switching cost).

Derivation. Identical algebra. Tilted Gibbs: $\pi^* \propto \pi_{\text{ref}} \exp(u^*/\beta)$. Hotz–Miller inversion recovers utility differences from CCPs. Substitute into BT likelihood. Fit π_θ by MLE.

Output. Estimated policy $\pi_{\hat{\theta}}(a \mid s)$.

Interpretation. The “structural policy function” - a behavioral rule whose log-ratio scores explain choice data under a RUM. Implicitly encodes utility. Called “CCP inversion” or “Hotz–Miller approach.”

LLM alignment	Structural econometrics	Mathematical object
Prompt x	State s	Element of $\mathcal{X}$
Completion y	Action a	Element of $\mathcal{A}(x)$
Reward $r^*(x, y)$	Utility $u^*(s, a)$	Latent score $f : \mathcal{X} \times \mathcal{A} \rightarrow \mathbb{R}$
Pretrained LLM π_{ref}	Default behavior	Reference measure on $\mathcal{A}$
KL penalty β	Rational inattention cost β	Regularization / noise scale
DPO loss	BT log-likelihood	$\sum \log \sigma(\beta \Delta \log(\pi_\theta / \pi_{\text{ref}}))$
“Aligned policy”	“Structural policy function”	MLE under BT with log-ratio scores
“Bypass reward model”	“CCP inversion”	Avoid fixed-point by inverting CCPs

What each field brings. The LLM literature brings scalable implementation (transformer architectures, LoRA, distributed training) and rich sequential structure (soft Bellman equations for chain-of-thought). The economics literature brings identification theory (what can and cannot be recovered from data), welfare analysis (counterfactual policy evaluation), and a century of discrete choice econometrics (misspecification diagnostics, heterogeneity models, dynamic extensions).

C. Dictionary: Structural Economics $\leftrightarrow$ Preference Fine-Tuning

Structural economics	Preference-based fine-tuning
Context / environment x	Prompt / task x
Alternative y	Candidate completion y
Utility $u(x, y)$	Reward / score $r(x, y)$
Random utility shocks (ε)	Rater noise / preference randomness
Logit / BT likelihood	Pairwise preference loss (DPO)
CCP inversion (Hotz–Miller)	Log-ratio identity (Corollary 1)
Entropy / information cost	KL divergence to reference policy

This dictionary is the backbone for specializing to cultural alignment: “culture” becomes either a parameter in $r(x, y; z)$ or a shifter of the state distribution.

D. Worked Micro-Example: Finite Candidate Set

Fix a context x and three candidates $\mathcal{Y}(x) = \{a, b, c\}$. Suppose a reference policy is $\pi_{\text{ref}}(\cdot | x) = (0.6, 0.3, 0.1)$ and the reward levels are $r^*(x, \cdot) = (0, 1, 2)$ with $\beta = 1$.

The optimal policy is $\pi^*(y | x) \propto \pi_{\text{ref}}(y | x) e^{r^*(x, y)}$.

Compute unnormalized weights:

$$w_a = 0.6 \cdot e^0 = 0.6, \quad w_b = 0.3 \cdot e^1 \approx 0.815, \quad w_c = 0.1 \cdot e^2 \approx 0.739.$$

Normalizing: $\pi^* \approx (0.28, 0.38, 0.34)$.

BT probability that c beats a :

$$\sigma(r_c - r_a) = \sigma(2) \approx 0.88.$$

Verification via log-ratios:

$$\beta \left(\log \frac{\pi^*(c)}{\pi_{\text{ref}}(c)} - \log \frac{\pi^*(a)}{\pi_{\text{ref}}(a)} \right) = \log \frac{0.34}{0.1} - \log \frac{0.28}{0.6} \approx 1.22 - (-0.76) = 1.98,$$

which matches $\Delta r = 2$ up to rounding. Substitution reproduces the same BT probability. This is the reparameterization equivalence in arithmetic form.

What this example illustrates:

1. The reference policy matters: candidate c has the highest reward but the lowest reference probability, so its share in π^* is constrained by the KL cost.
2. The partition function $Z(x)$ cancels in all pairwise comparisons.
3. The log-ratio scores $\log(\pi^*(y)/\pi_{\text{ref}}(y))$ encode the same information as the reward differences.
4. The finite-candidate setting is the cleanest environment for the proof; in practice, DPO operates on sampled candidate sets.

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